DIVIDED POWER DIFFERENTIALS VIA THE COPRODUCT IN CRIS(C/A)

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file crystalline.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original reference 07MD: sources/stacks-chow/tags.txt, source lines 9723–9723. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 07HT: sources/stacks-chow/tags.txt, source lines 9563–9563. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 07MD. Let (B, J, δ) be an object of CRIS(C/A). Let $(B(1), J(1), \delta(1))$ be the coproduct of (B, J, δ) with itself in CRIS(C/A). Denote $K = \mathrm{Ker}(B(1) \to B)$. Then $K \cap J(1) \subset J(1)$ is preserved by the divided power structure and*

$$\Omega_{B/A,\delta} = K/\left(K^2 + (K \cap J(1))^{[2]}\right)$$

canonically.

Proof. Word for word the same as the proof of Lemma 07HT. The only point that has to be checked is that the divided power ring $D = B \oplus M$ is an object of CRIS(C/A) and that the two maps $B \to D$ are morphisms of CRIS(C/A). Since $D/(J \oplus M) = B/J$ we can use $C \to B/J$ to view D as an object of CRIS(C/A) and the statement on morphisms is clear from the construction. □

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